

Proxy Service

A comprehensive, containerized proxy service with VPN routing, intelligent caching, and network-wide access for all devices.

Features

- **HTTP/HTTPS Proxy:** Squid-based caching proxy for web traffic
- **SOCKS5 Proxy:** Dante SOCKS proxy for flexible protocol support
- **VPN Routing:** Route all proxy traffic through OpenVPN for privacy
- **Intelligent Caching:** Reduce bandwidth by caching frequently accessed content
- **Streaming Cache:** Special handling for video/audio streaming services
- **Network-Wide Access:** Share proxy connection with all devices on your network
- **Auto-Recovery:** Automatic VPN reconnection and service health monitoring
- **Cache Invalidation:** Automatic and manual cache cleanup mechanisms

Quick Start

```
# 1. Clone the repository
git clone git@github.com:vasic-digital/Proxy.git
cd Proxy

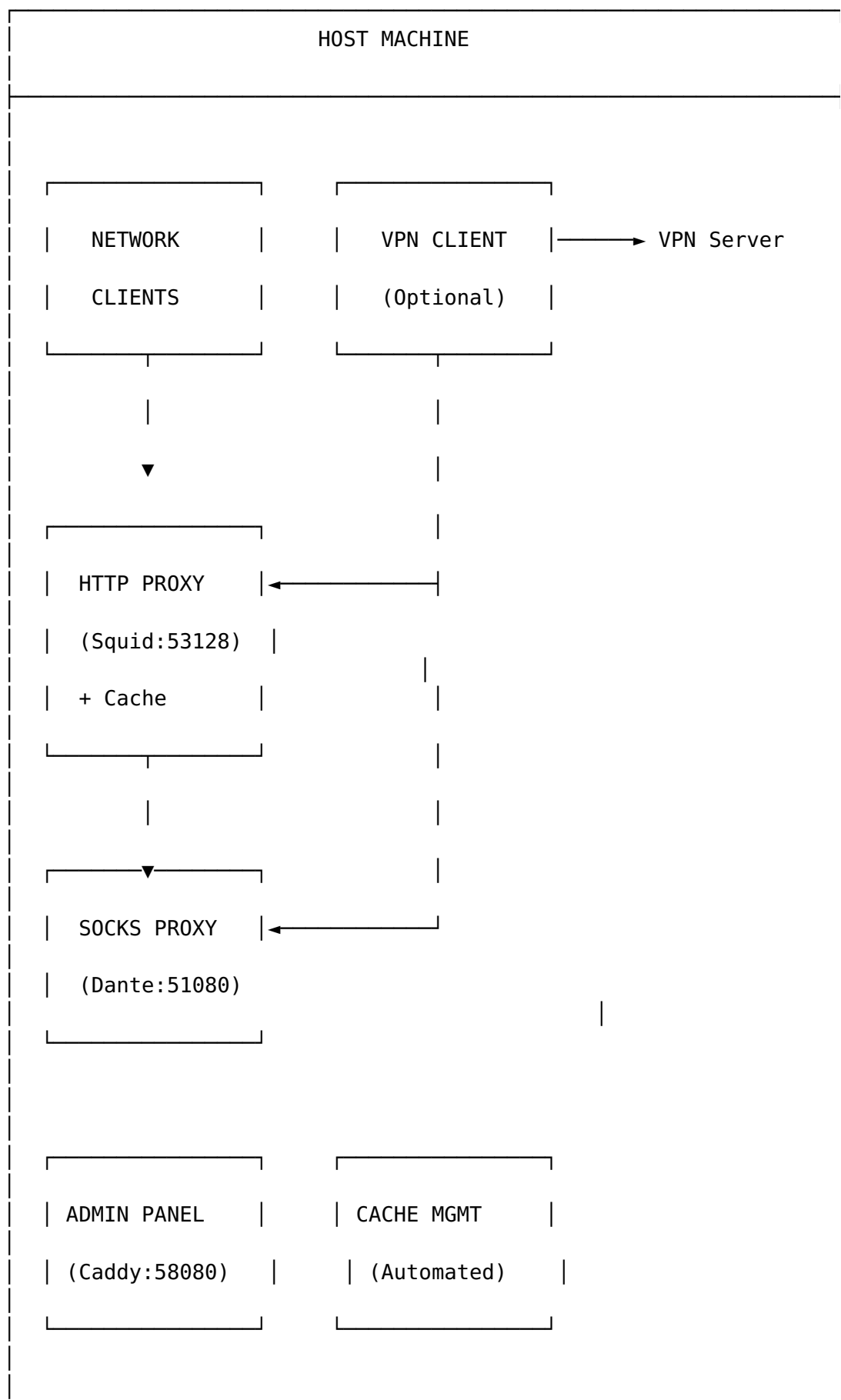
# 2. Copy and configure environment
cp .env.example .env
# Edit .env with your settings

# 3. Initialize the service
./init

# 4. Start the proxy
./start

# 5. Check status
./status
```

Architecture



Directory Structure

```
Proxy/
├── .env                                # Environment configuration (git-
ignored)
├── .env.example                       # Configuration template
├── .gitignore                         # Git ignore rules
├── AGENTS.md                          # AI agent guidelines
├── README.md                          # This file
├── USER_GUIDE.md                     # Detailed user manual
├── docker-compose.yml                 # Container service definitions
├── init                               # Environment initialization script
├── start                              # Start services
├── stop                                # Stop services
├── restart                            # Restart services
├── status                             # Service status checker
├── cache                             # Cache management script
├── lib/
│   └── container-runtime.sh          # Shared runtime functions
├── config/
│   ├── squid/
│   │   └── squid.conf                # Squid proxy configuration
│   ├── dante/
│   │   └── sockd.conf                # SOCKS proxy configuration
│   ├── caddy/
│   │   └── Caddyfile                 # Admin interface config
│   └── streaming.conf                # Streaming cache settings
├── scripts/
│   ├── cache-invalidator.sh          # Cache cleanup automation
│   └── vpn-monitor.sh                # VPN health monitoring
├── services/
│   └── admin/
│       └── index.html                 # Admin panel interface
├── docs/
│   ├── ARCHITECTURE.md               # System architecture
│   ├── CACHE.md                      # Caching documentation
│   ├── VPN.md                        # VPN configuration guide
│   ├── NETWORK_MODES.md              # Network mode comparison
│   ├── TROUBLESHOOTING.md            # Common issues and solutions
│   ├── CLIENT_SETUP.md               # Client setup for all devices
│   ├── ANDROID_TV.md                 # Android TV setup guide
│   ├── BROWSERS.md                   # Browser configuration
│   └── MOBILE_DEVICES.md              # iOS/Android setup
├── tests/
│   └── run-tests.sh                  # Test runner
├── logs/                             # Log files (git-ignored)
```

```
└─ Upstreams/
   └─ GitHub.sh # Git upstream configuration
```

Configuration

Environment Variables

Variable	Description	Default
CONTAINER_RUNTIME	Runtime to use (podman/docker/auto)	auto
HTTP_PROXY_PORT	HTTP/HTTPS proxy port	53128
SOCKS_PROXY_PORT	SOCKS5 proxy port	51080
PROXY_ADMIN_PORT	Admin panel port	58080
USE_VPN	Enable VPN routing	false
VPN_USERNAME	VPN provider username	-
VPN_PASSWORD	VPN provider password	-
VPN_OVPN_PATH	Path to .ovpn config file	-
CACHE_DIR	Cache storage directory	./cache
CACHE_MAX_SIZE_GB	Maximum cache size	50
CACHE_MAX_AGE_DAYS	Max age before invalidation	30

See `.env.example` for complete configuration options.

Usage

Starting the Service

```
# Start with VPN (if configured)
./start

# Start without VPN
./start --no-vpn

# Start with verbose output
./start -v

# Pull latest images before starting
./start --pull
```

Stopping the Service

```
# Stop services
./stop
```

```
# Stop and remove containers
./stop --remove

# Stop, remove containers and images
./stop --purge

# Stop and clear cache
./stop --clean-cache
```

Checking Status

```
# Basic status
./status

# Detailed status
./status -v

# JSON output
./status --json

# Watch mode (continuous monitoring)
./status --watch
```

Managing Cache

```
# Show cache statistics
./cachectl stats

# Clear all cache
./cachectl clear

# Force clear without confirmation
./cachectl clear -f

# Run invalidation (remove stale files)
./cachectl invalidate

# Trim cache to specific size
./cachectl trim 30 # Trim to 30GB
```

Client Configuration

Linux/macOS

```
# Set environment variables
export HTTP_PROXY="http://HOST_IP:53128"
export HTTPS_PROXY="http://HOST_IP:53128"
export ALL_PROXY="socks5://HOST_IP:51080"
```

```
export NO_PROXY="localhost,127.0.0.1"
```

Or add to ~/.bashrc or ~/.zshrc for persistence

System-wide (Linux)

Add to /etc/environment

```
HTTP_PROXY="http://HOST_IP:53128"  
HTTPS_PROXY="http://HOST_IP:53128"  
NO_PROXY="localhost,127.0.0.1"
```

Windows

PowerShell

```
$env:HTTP_PROXY = "http://HOST_IP:53128"  
$env:HTTPS_PROXY = "http://HOST_IP:53128"
```

Command Prompt

```
set HTTP_PROXY=http://HOST_IP:53128  
set HTTPS_PROXY=http://HOST_IP:53128
```

Browser Configuration

Firefox

1. Settings → General → Network Settings
2. Select "Manual proxy configuration"
3. HTTP Proxy: HOST_IP, Port: 53128
4. SOCKS Host: HOST_IP, Port: 51080, SOCKS v5

Chrome/Edge

Use system proxy settings or extensions like SwitchyOmega.

VPN Configuration

1. Obtain your VPN provider's .ovpn configuration file
2. Set environment variables:

```
USE_VPN=true  
VPN_USERNAME=your_username  
VPN_PASSWORD=your_password  
VPN_OVPN_PATH=/path/to/config.ovpn
```

3. Start the service: ./start

VPN Features

- **Auto-Reconnect:** Automatically reconnects on disconnect
- **Health Monitoring:** Periodic connectivity checks
- **Cache Invalidation:** Optional cache clear on VPN reconnect

Caching

How It Works

1. **Request Interception:** Proxy intercepts HTTP/HTTPS requests
2. **Cache Lookup:** Checks if response is already cached
3. **Freshness Check:** Validates cache freshness
4. **Response:** Serves from cache or fetches from origin
5. **Storage:** Caches valid responses for future use

Streaming Cache

Special handling for video/audio streaming:

- Chunk-based caching for partial content
- Range request support
- Configurable streaming domains
- Separate cache pool

Cache Invalidation

Automatic invalidation:

- Files older than `CACHE_MAX_AGE_DAYS`
- When cache exceeds `CACHE_MAX_SIZE_GB`
- On VPN reconnect (if configured)

Manual invalidation:

```
./cachectl invalidate
```

Security

Best Practices

1. **Network Restrictions:** Configure `ALLOWED_NETWORKS` to limit access
2. **Authentication:** Enable `PROXY_AUTH_ENABLED` for user authentication
3. **Firewall:** Use host firewall to restrict access

4. **VPN:** Enable VPN for privacy and geo-restriction bypass

Firewall Configuration

Allow from specific network only

```
sudo ufw allow from 192.168.1.0/24 to any port 53128
```

```
sudo ufw allow from 192.168.1.0/24 to any port 51080
```

Troubleshooting

Service Won't Start

1. Check logs: `./logs/proxy.log`
2. Verify ports are not in use: `ss -tuln | grep -E '53128|51080'`
3. Check container runtime: `./init --check`

VPN Not Connecting

1. Verify VPN credentials in `.env`
2. Check `.ovpn` file path and permissions
3. Check VPN container logs: `$COMPOSE_CMD logs proxy-vpn`

Cache Not Working

1. Verify cache directory exists and is writable
2. Check Squid logs: `./logs/squid/cache.log`
3. Verify cache configuration: `./cachectl stats`

Connection Refused

1. Verify service is running: `./status`
2. Check firewall rules
3. Verify client is using correct IP and port

Development

Running Tests

```
./tests/run-tests.sh
```

Building Custom Images

```
podman build -t proxy-custom:latest .
```

Contributing

See [CONTRIBUTING.md](#) for contribution guidelines.

License

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Support

- **Issues:** [GitHub Issues](#)
- **Documentation:** See docs/ directory for detailed documentation
 - [Client Setup Guide](#) — Configure any device
 - [Android TV Guide](#) — Android TV / Google TV setup
 - [Browser Guide](#) — Firefox, Chrome, Edge, Safari
 - [Mobile Devices](#) — iOS and Android
 - [Network Modes](#) — VPN vs no-VPN explained
 - [VPN Setup](#) — VPN configuration
 - [Architecture](#) — System design
 - [Caching](#) — Cache management
 - [Troubleshooting](#) — Common issues

Network Modes Explained

Mode 1: Containerized VPN (USE_VPN=true)

Client → Proxy → VPN Container → VPN Server → Internet

- VPN runs inside a container
- Complete isolation
- Best for dedicated proxy server

Mode 2: Host VPN Pass-through (--host-vpn)

Client → Proxy (host network) → Host VPN → VPN Server → Internet

- Uses host's existing VPN connection
- Containers share host network namespace
- Best when host already has VPN

Mode 3: No VPN (--no-vpn or USE_VPN=false)

Client → Proxy → Direct Internet

- No VPN routing

- Bridge network isolation
- Best for local caching only

Quick Comparison

Mode	Command	VPN Source	Network
Containerized	<code>./start (with USE_VPN=true)</code>	Container	VPN container
Host Pass-through	<code>./start --host-vpn</code>	Host system	Host network
No VPN	<code>./start --no-vpn</code>	None	Bridge

Tracked-Items + Status Documents

Per Helix Constitution §11.4.57. Each row links the tracked status / continuation document with its current Revision and Last-modified stamp (auto-derived from each document's §11.4.44 header). The canonical Issues / Fixed trackers are not present in this project and are omitted rather than fabricated.

Document	Last modified	Revision	Markdown	HTML	PDF
Continuation	2026-07-02T00:00:00Z	12	md	html	pdf
Feature Status	2026-07-01T15:30:00Z	3	md	html	pdf
Feature Status Summary	2026-07-01T15:30:00Z	3	md	html	pdf
Hardening Status	2026-07-01T13:26:00Z	3	md	html	pdf
Hardening Status Summary	2026-07-01T13:26:00Z	3	md	html	pdf
Security Status	2026-07-01T14:48:00Z	3	md	html	pdf
Security Status Summary	2026-07-01T14:48:00Z	3	md	html	pdf
Let's Encrypt Status	2026-07-01T11:42:00Z	3	md	html	pdf
Let's Encrypt Status Summary	2026-07-01T11:42:00Z	3	md	html	pdf

Document	Last modified	Revision	Markdown	HTML	PDF
VPN-LAN Integration Status	2026-07-01T19:05:00Z	3	md	html	pdf
VPN-LAN Integration Status Summary	2026-07-01T19:05:00Z	3	md	html	pdf
VPN-LAN Feature Status	2026-07-01T19:15:00Z	2	md	html	pdf
VPN-LAN Feature Status Summary	2026-07-01T19:15:00Z	2	md	html	pdf